ELECTROMAGNETIC FLOWMETER

User Manual



Contents

1. Product Description 01

2. Flowmeter Menu Structure 02

3. Flowmeter Parameter Description 02

3.1. Flow Parameter Setting 02

3.2. Total Flow Setup 04

3.3. Calculation 05

3.4. Pulse Output Setup 05

3.5. Communication Setup 06

3.6. Factory Parameter Setup 06

3.7. Energy Parameter Setting 09

4. How to Setup the Parameter 10

5. Installation 12

1. Product Description

The Electromagnetic Flow Meter is a highly intelligent, maintenance-free flow measurement instrument designed to provide accurate, stable, reliable, and precise flow measurement. Based on advanced electromagnetic flow measurement technology, it offers excellent long-term stability and low operating costs.

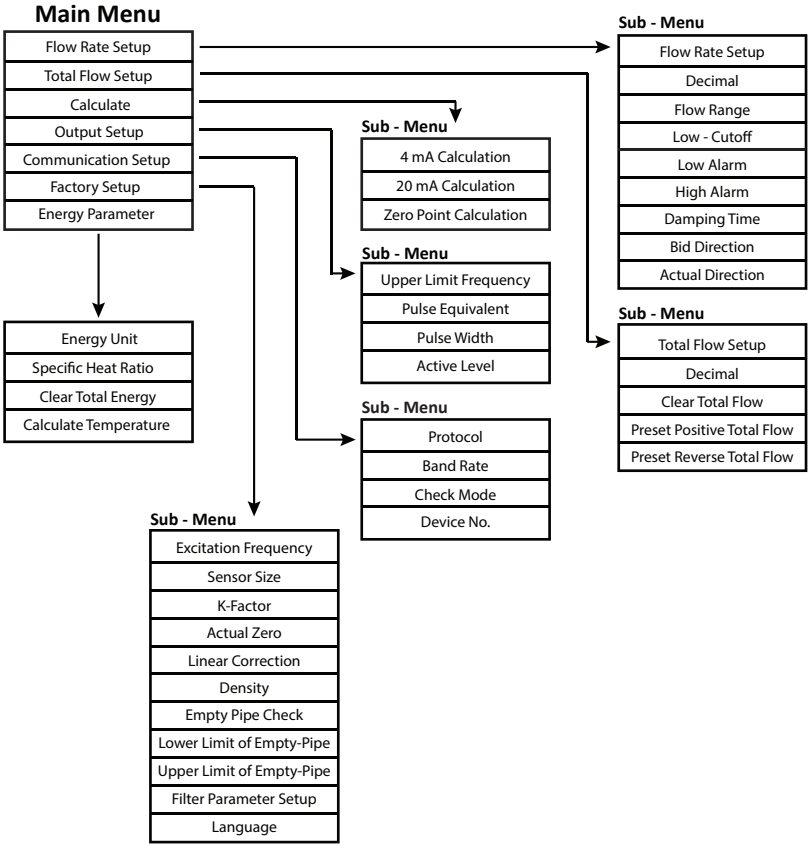
The Electromagnetic Flow Meter operates on Faraday's Law of Electromagnetic Induction and is suitable for measuring the flow of electrically conductive liquids. Since the measurement principle is independent of the liquid's viscosity, density, pressure, and temperature, the instrument provides a true volumetric flow measurement. The measured flow rate is directly proportional to the induced electrical signal generated within the measuring tube.

The flow meter is widely used across various process industries and utility applications, including:

- Petroleum and Oil & Gas Industries
- Chemical and Petrochemical Industries
- Steel and Metal Industries
- Food and Beverage Industries
- Power Generation
- Pulp and Paper Industries
- Water Supply and Drainage
- Water Conservancy
- Water Resource Management
- Industrial and Municipal Water Treatment
- Other applications involving electrically conductive liquids

Its robust construction, high measurement accuracy, low maintenance requirements, and reliable performance make it suitable for demanding industrial flow measurement applications.

2. Flowmeter Menu Structure



3. Flowmeter Parameter Description

3.1 Flow Parameter Setting

PV Unit	Option: L/s, L/m, L/h, m^3/s, m^3/m, m^3/h, USG/s, USG/h, Kg/s, Kg/m, Kg/h, t/s, t/m, t/h Default: m^3/h define the unit of the flow rate. L(liter), h(hour), t(ton), s(second), m(minute)
PV Decimal	Option: 0, 1, 2, 3 Default: 1 Define decimal point position of flow rate.
Flow Range	Flow Point: 99999999.00 - 0.00 m^3/h Default: 100.0 m^3/h When the instantaneous flow rate reaches this set value, the output current is 20mA. Change this parameter will affect: current output, high and low flow alarm, etc.
Low Flow Cut - off	Float Point: 9.90 ~ 0.00% Default: 0.0% This set value is a percentage of flow range.
Low Alarm	Float Point: 99.00 ~ 0.00% Default: 0.0% This value is a percentage of flow range, for example, if this value is setup to ten percent(10%) of flow range. If the absolute value of instantaneous flow < (flow range * 10%), then the transmitter will output the high alarm signal, the contact of high alarm will close.
High Alarm	Float Point: 99.00 ~ 1.00% Default: 90.0% This value is a percentage of flow range, for example, if this value is setup to ten(10), then equivalent to ten percent(10%) of flow range. If the Absolute value of instantaneous flow > (flow range * 10%), then transmitter will output the high alarm signal, the contact of high alarm will close.
Damping Time	Float Point: 30.0 ~ 0.1 Default: 0.1 **Note: for this type of transmitter, no need to set damping time, flow measurement is very stable. So you can keep the damp time is default "0.1".**

Bid Flow Direction	Option: Bid, Reverse, Positive Default: Bid When set to positive, the reverse flow rate will not displayed (display: 0) When set to reverse, the forward flow rate will not be displayed (display: 0) When set to Bi - directional, the positive and negative flow can be displayed.
Actual Flow Direction	Option: Reverse, Positive Default: Positive

3.2 Total Flow Setup

Define the relevant parameters of the total flow.

Total Flow Unit	Option: L(liter), m³, USG, Kg, t(ton) Default: m³ Define the total flow unit.
Decimal	Option: 0, 1, 2, 3 Default: No Define the decimal point bit of the total flow value.
Clear Total Flow	Option: No, Yes Default: No Clear the total flow. **Note: If the "Flow Over!" alarm information, please timely clear or preset the total flow to avoid affecting measurement.**
Positive Total Preset	Float Point: 9999999999 ~ 0.00 m³ Default: the current total flow After setting this value, the current total flow will be covered by this set of value.
Reverse Total Preset	Float Point: 9999999999 ~ 0.00 m³ Default: the current total flow After setting this value, the current total flow will be covered by this set of value.

3.3 Calculation

Adjust analog output and zero value of flow meter.

4mA Calculate	Float Point: 5.0 ~ 3.0 Default: 0.0 After you go to the item, please use the Precision current meter to measure the current output value. If this result is not 4.0mA, then input will automatically complete the calibration operation.

20mA Calculate	Option: 21.0 ~ 19.0 Default: 0.0 After you go to this item, please use the Precision current meter to measure the current output value. If this result is not 20.0mA, then input the actual value which measure to this position, then the instrument will automatically complete the calibration operation.
Calculate Zero	Option: No, Yes Default: No When the fluid is static in the pipeline and completely filled with the pipeline, the zero value equal to the voltage difference between the two signal electrodes of the electromagnetic flowmeter. **Note: After the implementation of zero calibration, if the zero value is high, please check the sensor.** **In general, the zero value can not exceed 2mV. If the zero value is more than 2mV, then the accuracy of the flow meter is not ok. If the zero value is high, but the sensor has no quality problem, you can reduce the excitation frequency to reduce the zero value.**

3.4 Pulse Output Setup

Setup the parameter of pulse output or frequency output.

Upper Frequency Limit	Float Point: 5000.0 - 100.0 Hz Default: 0.0 **Output Frequency(Hz) = Flow rate(m^3/h) + Flow range(m^3/h) * Upper frequency limit(Hz)** For Example, the flow rate is 100m^3/h, and the flow range is 200m3/h, and the “upper frequency limit” is set to 2000Hz, then this time, the output frequency is 1000Hz. **Note: If use the “Frequency output” then must set “Pulse equivalent = 0”.**
Pulse Equivalent	Option: 9999.0 - 0.0 Default: 0.0 When the pulse equivalent = 0.0, then output frequency signal When the pulse equivalent > 0.0, then pulse output is determined by the “Pulse Equivalent”. The “Pulse Equivalent” unit is “liter per one pulse”.
Pulse width (ms)	Floating Point: 1000.0 ~ 0.0 ms Default: 0.0 When this value is “0”, the duty cycle of the output pulse is 1:1
Active Level	Option: Low (Active Low), High (Active High) Default: Low (Active Low) This parameter is related to the “Pulse Width”.

	When set to "**Low**", the set value of the pulse width is used as the pulse width of the pulse output low level. When set to "**High**", the set value of the pulse width is used as the pulse width of the pulse output high level. In general, this parameter does not need to be set.

3.5 Communication Setup

Setup RS485 communication parameter.

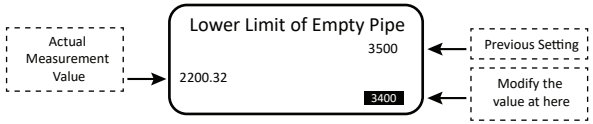
Communication Protocol	Option: Modbus-RTU, Modbus-ASCII Default: Modbus-RTU
Baud Rate	Option: 1200, 2400, 4800, 9600, 19200, 38400 Default: 19200 **Note: Please see the baud rate >= 9600**
Data Bit	Option: 7, 8 Default: 8 **Note: If use RTU protocol, prohibit to choose "7".**
Check Mode	Option: None, Odd, Even Default: Even
Device No.	Option: 247 ~ 1 Default: 1

3.6 Factory Parameter Setup

Password 052500. Setup the key parameters.

Excitation Frequency	Option: 25Hz, 12.5Hz, 6.25Hz, 3.125Hz Default: 6.25Hz If the sensor size is greater than DN500, select 3.125Hz. If the sensor size is less than DN500, select 6.25Hz. If the inductance of the sensor is very small, you can choose 25Hz. If the zero value is high, but the sensor has no quality problem, you can reduce the excitation frequency to reduce the zero value. If the flow rate is not very stable, you can try to reduce the excitation frequency.
Sensor Size	Option: 1, 2, 3, 6, 8, 10, 15, 20, 25, 32, 40, 50, 65, 80, 100, 125, 150, 200, 250, 300, 350, 400, 450, 500, 550, 600, 650, 700, 800, 900, 1000, 1100, 1200, 1300, 1400, 1500, 1600, 1700, 1800, 2000, 2200, 2400, 2600, 2800, 3000 mm Default: 50

K - Factor	Floating Point: 9.9000 ~ 0.0100 Default: 0.1 This Parameter is determined when the real flow calibration is carried out. This parameter is only related to sensor, which indicates the characteristics value of the sensor. **New Sensor K = the old Sensor K X (Flow rate the calibrated flowmeter / actual flow rate).** **After you finished the test by water, then need setup the final K-Factor at here.**
Actual Zero	Default: 0.0 Unit: mV (voltage) When the fluid is still in the pipeline and completely filled with the pipeline, the zero value is equal to the voltage difference between the two electrodes of the electromagnetic flowmeter. Note: After the implementation of zero calibration, if the zero value is high, please check the sensor. In general, the zero value can not exceed 2mV. If the zero value is more than 2mv, then the accuracy of the flowmeter is not ok.
Linear Correction	The correction range: The default value is 0 ~ 2m/s (refers to the scope of the role of non linear correction). Default: 0.0mV
Density	Floating point: 999999 ~ 0.1 kg/m³ Default: 1000.0 kg/m³ When you need to use the quality unit, set the exact density. If the unit is Volume Unit, then this item have no any affect.
Empty Pipe Check	Option: Enable or Prohibit Default: Enable If you select “prohibit”, then the flowmeter transmitter does not carry out empty - pipe detection, then even if there is no water, it may be show the flow rate.
Lower Limit of empty pipe (as shown in the picture)	Float point Default: 3500 **DTI type EMF transmitter can measure the resistance between electrodes automatically. For three electrode flowmeter or four electrodes flow meter or the flowmeter with ground ring, if the pipe is fully filled with water, then the resistance value is about the 2000, because of the different conductivity of the fluid, this value may be somewhat different, and for separate type flow meters, the length of the cable also affects this value. In fact, for most of the fluid, this value is basically between 2000 - 3000 when the fluid filled pipe. And when the pipe is empty (no water), then the value will become to about 4000 - 4500, so we setup the “Lower limit of empty-pipe” to 3000. And this is type value.** **If you setup the “Lower limit of empty-pipe” to 3500, the meaning is that the**

	Floating Point: 9.9000 ~ 0.0100 Default: 0.1 This Parameter is determined when the real flow calibration is carried out. This parameter is only related to sensor, which indicates the characteristics value of the sensor. **New Sensor K = the old Sensor K X (Flow rate the calibrated flowmeter / actual flow rate).** **After you finished the test by water, then need setup the final K-Factor at here.**
	
Upper Limit of Empty Pipe	Float point Default: 5500 **DTI type EMF transmitter can measure the resistance between electrodes automatically. For two electrode flowmeter or the flow meter which is not well connected to ground, if the pipe is fully filled with water, then the resistance value may be >6000, because of the different conductivity of the fluid, this value may be somewhat different, and for separate type flow meters, the length of the cable also affects this value.** **And when the pipe is empty (no water), then the value will become to about 4000 to 4500. So we setup the "upper limit of empty-pipe" to 5500. And this is type value.** **If you setup the "upper limit of empty-pipe" to 5500, the meaning is that the flowmeter determines the current state to "empty pipe (no water in pipe)" when the actual measurement value <5500.** **If the actual measurement value >5500, then the flowmeter determines that the pipe is filled with liquid.** **The actual measurement value is appeared in the left of this menu. In some cases it may be possible to change this setting.** **For all electromagnetic flowmeters, whether it is the two electrode or three electrode or with ground ring, DTI Transmitter determines the "empty pipe (no water in pipe)" according to actual measurement value, "upper limit empty-pipe" and "lower limit empty-pipe".** **If "lower limit empty-pipe" ≤ actual value ≤ "upper limit empty-pipe", then the flowmeter determines to "empty pipe (no water in pipe)".** **If actual value < "lower limit empty-pipe", then "full-pipe (filled water)".** **If actual value > "upper limit empty-pipe", then also "full-pipe".**

Filter-Parameter Setup	Filter time Option: 0, 1, 2 Default: 0 Note: In the normal state, the filtering time is “0”, and meets the requirements of the corresponding time for 0.3 seconds. If the flowrate is very unstable, and the sensor is ok, and this instability condition is caused by the fluid, then you can change the “filter time” to “1”. Filtering threshold: Option: 1, 2, 3, 4, 5 Default = 1

3.7 Energy Parameter Setup

Define the relevant parameters of the energy meter, and please note this menu will appear when you purchase the energy meter.

Energy Unit	Option: GJ/H, Kc/H, KW Default: GJ/H
Heat Capacity Ratio	Float point: 0.000 – 10.0 Default = 4.2
Correction Factor	Default = 1.0
Clear Total Energy	Option: No / Yes, Default = No If select the “Yes”, then clear the total energy value to zero
Calculate Temperature	The temperature measuring circuit of the converter is calibrated by using high precision resistance box.

4. How to Setup the Parameter

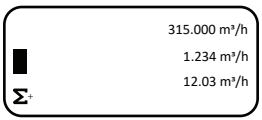


Figure 1: Flow Rate display Interface

- Press to enter the parameter setup menu, as shown in **Figure 2.**

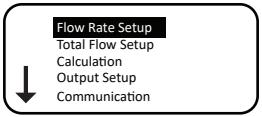


Figure 2.

- After the interface is shown in **Figure 2**, press or to choose different set items.
- Press then return to the flow display interface (Figure 1).
- Press or to move the cursor to the sub-menu. Press to go to the sub-menu to set up the parameter.

  For example, we need to set up the “Flowrate Parameter”. When this item becomes highlighted, press , then the display will show the menu as shown in **Figure 3**.

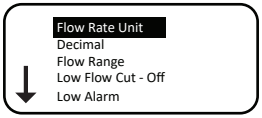


Figure 3.

- Press or to select the item which you want to modify. The selected item will brighten.
- If you need to return to the menu as shown in **Figure 2**, then press
- If you need to enter the next level of items, then press to set up the parameters as shown in **Figure 4**.

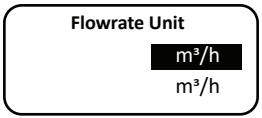


Figure 4.

- In this case, press or need to change the flowrate unit from "m³/h" to "m³/m", then press . The flowrate unit will become "m³/m", as shown in **Figure 5**.

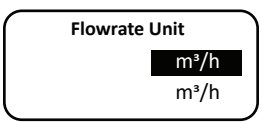


Figure 5.

- After you modify the parameter, if you need to save, then press . The system will prompt the **"confirmation"** and **"exit"** option, as shown in **Figure 6**.

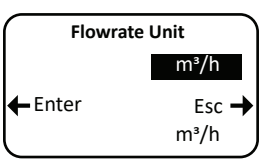


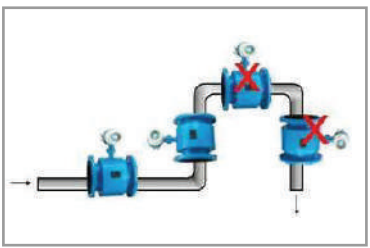
Figure 6.

- In this case, press , then save the setup and back (as shown in figure 3). If do not need the save the parameter, then press to back (as shown in **figure 3**).

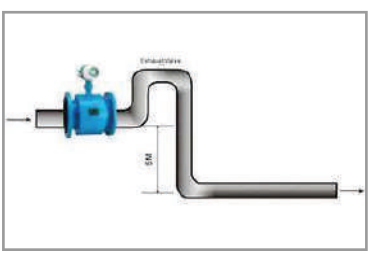
5. Installation

1. Install at the lowest point and vertical upward direction.

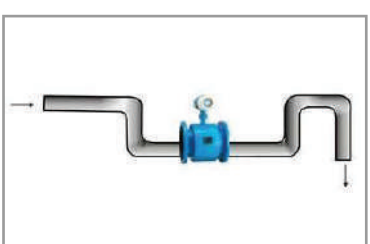
   Don't install at the highest point or vertical downward direction.



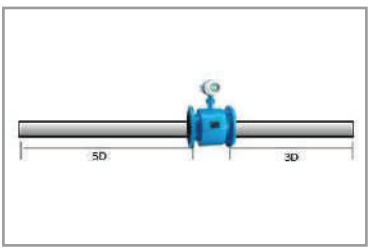
2. When drop is more than 5m, install exhaust valve at the downstream.



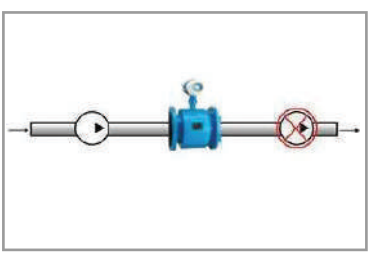
3. Install at the lowest point when used in open drain pipe.



4. Need 5D of upstream and 3D of downstream.



5. Don't install it at the entrance of pump, install it at the exit of pump.



6. Install at the rising direction.

